List Of Standard Specifications

Standard	Specifications		Yield strength N/mm²			Tensile Strength N/mm²	Elongation min. %	Charpy V-notch Temp. (°C) Energy (J)				
			t≤16mm	16<t≤40	t>40mm			20	0	-20	-40	-50
Contd.							Contd. PRESSURE VESSELS, GENERAL					
EN 10028-6												-60
	P460	Q, QH						60	40	27	-	-
		QL1	[8] 460	[8]* 440	[8]** 400	[8]*** 500-720	19	-	60	40	27	-
		QL2						-	80	60	40	27
	P500	Q, QH						60	40	27	-	-
		QL1	[8] 500	[8]* 480	[8]** 440	[8]*** 590-770	17	-	60	40	27	-
		QL2						-	80	60	40	27
	P690	Q, QH						60	40	27	-	-
		QL1	[8] 690	[8]* 670	[8]** 630	[8]*** 770-940	14	-	60	40	27	-
		QL2						-	80	60	40	27
							PRESSURE VESSELS, LOW TEMPERATURE SERVICE					
EN 10207							3-40mm / 40-60					
	P235S		235	225	215	360-480	26 / 25	-	-	28	-	-
	P265S		265	255	245	410-530	22 / 22	-	-	28	-	-
	P275SL		275	265	255	390-510	24 / 24	-	-	-	-	28

Note:

1) For t≤50mm

1)* For 50<t≤100mm

1)** For 100<t≤150mm. For t>150mm, see EN 10137-2.

1)*** For t≤100mm. For t>100mm, see EN 10137-2.

2) For t≤50mm.

2)* For 50<t≤70mm.

3) For LR and ABS plates the difference between A, B, D, E, F grades are the impact tests. They are made at the following temperatures:

A grade	+20 °C	E/EH grade	-40°C
B/AH grade	0°C	FH grade	-60°C
D/DH grade	-20°C		

4) For plates thicker than 60mm, see EN 10028-2:1992.

5) For plates thicker than 50mm, see EN 10028-3:1992.

5)* For plates thicker than 70mm, see EN 10028-3:1992.

6) For t≤30mm.

6)* For 30<t≤50mm.

7) Maximum thickness 63mm.

8) For t≤50mm.

8)* For 50<t≤100mm.

8)** For 100<t<150mm

8)*** For t≤100mm. For100<t≤150mm, see EN 10028-6.

List Of Standard Specifications

Standard	Specifications	Yield strength N/mm²			Tensile Strength N/mm²	Elongation min. %	Charpy V-notch Temp. (°C) Energy (J)				
		t≤16mm	16<t≤40	t>40mm			20	0	-20	-40	-50
PRESSURE VESSELS, GENERAL											
ASTM A516						50mm-200mm					
	Grade 55	min. 205			380-515	27-23	-	-	-	-	-
	Grade 60	min. 220			415-550	25-21	-	-	-	-	-
	Grade 65	min. 240			450-585	23-19	-	-	-	-	-
	Grade 70	min. 260			485-620	21-17	-	-	-	-	-
EN 10028-2	P235GH	235	225	[4] 215	[4] 360-480	25	-	27	-	-	-
	P265GH	265	255	[4] 245	[4] 410-530	23	-	27	-	-	-
	P295GH	295	290	[4] 285	[4] 460-580	22	-	27	-	-	-
	P355GH	355	345	[4] 335	[4] 510-650	21	-	27	-	-	-
	16 Mo 3	275	270	[4] 260	[4] 440-590	24-23	31	-	-	-	-
	13 CrMo 4-5	300	[4] 295		[4] 450-600	20	31	-	-	-	-
	10 CrMo 9-10	310	300	[4] 290	[4] 480-630	18	31	-	-	-	-
	11 CrMo 9-10	min. 310			520-670	18	31	-	-	-	-
EN 10028-3						≤70mm / 70-150					
	P275 N, NH						55	47	40	-	-
	NL1	275	275	[5] 265	[5]* 390-510	24 / 23	63	55	47	34	27
	NL2						100	90	65	40	30
	P355 N, NH						55	47	40	-	-
	NL1	355	355	[5] 345	[5]* 490-630	22 / 21	63	55	47	34	27
	NL2						100	90	65	40	30
	P460 N, NH						55	47	40	-	-
	NL1	460	450	[5] 440	[5]* 570-720	17 / 16	63	55	47	34	27
	NL2						100	90	65	40	30
EN 10028-4		t<30mm		30mm<t≤50mm		t<50mm					
	11 MnNi 5-3	[6] 285		[6]* 275	420-530	24	70	60	55	50	45
	13 MnNi 6-3	[6] 355		[6]* 345	490-610	22	70	60	55	50	45
	15 MnNi 6	[6] 355		[6]* 345	490-640	22	65	65	65	60	50
	12 Ni 14	[6] 355		[6]* 345	490-640	22	65	60	55	55	50
	12 Ni 19	[6] 390		[6]* 380	530-710	20	70	70	70	65	65
	X8 Ni9 HT 640	[6] 490		[6]* 480	640-840	18	70	70	70	70	70
	X8 Ni9 HT 680	[6] 585		[6]* 575	680-820	18	120	120	120	120	120
	X7 Ni 9	[6] 585		[6]* 575	680-820	18	120	120	120	120	120
EN 10028-5	P355 M						60	40	27	-	-
	ML1	355	355	[7] 345	450-610	22	-	60	40	27	-
	ML2						-	80	60	40	27
	P420 M						60	40	27	-	-
	ML1	420	400	[7] 390	500-660	19	-	60	40	27	-
	ML2						-	80	60	40	27
	P460 M						60	40	27	-	-
	ML1	460	440	[7] 430	530-720	17	-	60	40	27	-
	ML2						-	80	60	40	27
EN 10028-6											-60
	P355 Q, QH						60	40	27	-	-
	QL1	[8] 355	[8] 335	[8]** 315	[8]*** 490-630	22	-	60	40	27	-
	QL2						-	80	60	40	27

List Of Standard Specifications

Standard	Specifications	Yield strength N/mm²			Tensile Strength	Elongation min. %	Charpy V-notch Temp. (°C) Energy (J)				
		t≤16mm	16<t≤40	t>40mm	N/mm²	L_o=5.65$\sqrt{S_o}$	20	0	-20	-40	-50
BRIDGES, FLOOD GATES, STORAGE TANKS, WATER TANKS, BUILDINGS, CRANE STRUCTURES											
EN 10025-6								0	-20	-40	
	S460Q, QL, QL1	[1] 460	[1]* 440	[1]** 400	[1]*** 550-720	17		40/50/60	30/40/50	-/30/40	
	S500Q, QL, QL1	[1] 500	[1]* 480	[1]** 440	[1]*** 590-770	17		40/50/60	30/40/50	-/30/40	
	S550Q, QL, QL1	[1] 550	[1]* 530	[1]** 490	[1]*** 640-820	16		40/50/60	30/40/50	-/30/40	
	S620Q, QL, QL1	[1] 620	[1]* 580	[1]** 560	[1]*** 700-890	15		40/50/60	30/40/50	-/30/40	
	S690Q, QL, QL1	[1] 690	[1]* 650	[1]** 830	[1]*** 760-940	14		40/50/60	30/40/50	-/30/40	
	S890Q, QL, QL1	[1] 890	[1]* 830	-	[1]*** 880-1100	11		40/50/60	30/40/50	-/30/40	
	S960Q, QL, QL1	[1] 960	-	-	[1]*** 980-1150	10		40/50/60	30/40/50	-/30/40	
EN 10137-3		3mm<t≤50mm		50mm<t≤70mm				0	-20	-40	
	S500A, AL	[2] 500		[2]* 480	600-700	17		55/65	40/50	-/40	
	S550A, AL	[2] 550		[2]* 530	650-820	16		55/65	40/50	-/40	
	S620A, AL	[2] 620		[2]* 580	710-880	15		55/65	40/50	-/40	
	S690A, AL	[2] 690		[2]* 650	760-930	14		55/65	40/50	-/40	
SPECIAL STEEL											
BS 7191	275D	275	265	-	430-580	22	-	-	40	-	-
	275E	275	265	265	430-580	22	-	-	-	40	-
	275EZ	275	265	265	430-580	22	-	-	-	40	-
	355D	355	345	-	490-640	20	-	-	50	-	-
	355E	355	345	-	490-640	20	-	-	-	50	-
	355EM	355	345	340	460-620	20	-	-	-	50	-
	355EMZ	355	345	340	460-620	20	-	-	-	50	-
	450EM	450	415	-	550-700	19	-	-	-	60	-
	450EMZ	450	415	-	550-700	19	-	-	-	60	-
HULL STRUCTURES											
ABS [3]							For thickness t ≤ 50mm				
	Grade A, B, D, E	min. 235			400-550	22	-	27	27	27	-60
	Gr. AH, DH, EH, FH 32	min. 315			440-590	22	-	34	34	34	34
	Gr. AH, DH, EH, FH 36	min. 355			490-620	21	-	34	34	34	34
	Gr. AH, DH, EH, FH 40	min. 390			510-650	20	-	41	41	41	41
LR [3]	Grade A, B, D, E	min. 235			400-520	22	27	27	27	27	-60
	Gr. AH, DH, EH, FH 32	min. 315			440-590	22	-	31	31	31	31
	Gr. AH, DH, EH, FH 36	min. 355			490-620	21	-	34	34	34	34
	Gr. AH, DH, EH, FH 40	min. 390			510-650	20	-	41	41	41	41
PRESSURE VESSELS, GENERAL											
ASTM A285						50mm-200mm	-	-	-	-	-
	Grade A	min. 165			310-450	30-27	-	-	-	-	-
	Grade B	min. 185			345-485	28-25	-	-	-	-	-
	Grade C	min. 205			380-515	27-23	-	-	-	-	-

List Of Standard Specifications

Standard	Specifications	Yield strength N/mm²			Tensile Strength	Elongation min. %	Charpy V-notch Temp. (°C) Energy (J)				
		t≤16mm	16<t≤40	t>40mm	N/mm²	L_o=5.65√S_o	20	0	-20	-40	-50
GENERAL STRUCTURES											
ASTM	*A36*	min. 250			400-550	50mm-200mm 23-20	-	-	-	-	-
ASTM A283						50mm-200mm					
	Grade A	min. 165			310-415	30-27	-	-	-	-	-
	Grade B	min. 185			345-450	28-25	-	-	-	-	-
	Grade C	min. 205			380-515	25-22	-	-	-	-	-
	Grade D	min. 230			415-550	23-20	-	-	-	-	-
ASTM A572						50mm-200mm					
	Grade 42	min. 290			415	24-20	-	-	-	-	-
	Grade 50	min. 345			450	21-18	-	-	-	-	-
	Grade 60	min.415			520	18-16	-	-	-	-	-
	Grade 65	min. 450			550	17-15	-	-	-	-	-
EN 10025-3&4 : 2004	S275N	min. 275			370-510		-	-	40	-	27
	S355N	min. 355			470-630		-	-	40	-	27
	S420N	min. 420			520-680		-	-	40	-	27
	S460N	min. 460			550-720		-	-	40	-	27
	S275M	min. 275			360-510		-	-	40	-	27
	S355M	min. 355			450-610		-	-	40	-	27
	S420M	min. 420			500-660		-	-	40	-	27
	S460M	min. 460			530-720		-	-	40	-	27
EN 10149-2	S315MC	min. 315			390-510	24	-	-	40	-	-
	S355MC	min. 355			430-550	23	-	-	40	-	-
	S420MC	min. 420			480-620	19	-	-	40	-	-
	S460MC	min. 460			520-670	17	-	-	40	-	-
	S500MC	min. 500			550-700	14	-	-	40	-	-
	S550MC	min. 550			600-760	14	-	-	40	-	-
	S600MC	min. 600			650-820	13	-	-	40	-	-
	S650MC	650 (t≤8mm)		620	700-880	12	-	-	40	-	-
	S700MC	700 (t≤8mm)		680	750-950	12	-	-	40	-	-
EN 10149-3	S260NC	min. 260			370-490	30	-	-	40	-	-
	S315NC	min. 315			430-550	27	-	-	40	-	-
	S355NC	min. 355			470-610	25	-	-	40	-	-
	S420NC	min. 420			530-670	23	-	-	40	-	-
JIS G 3101	SS330	205	195	175	330-430	28-21	-	-	-	-	-
	SS400	245	235	215	400-510	17-23	-	-	-	-	-
	SS490	285	275	255	490-610	15-21	-	-	-	-	-
	SS540	400	390	-	min. 540	13-17	-	-	-	-	-
JIS G 3106							0 °C		-5°C		
	SM400A, B, C	245	235	215	400-510	18-24	B/C 27/47		-	-	-
	SM490A, B, C	325	315	295	490-610	17-23	B/C 27/47		-	-	-
	SM490YA, YB	365	355	335	490-610	15-21	YB27		-	-	-
	SM520B, C	365	355	335	520-640	15-21	B/C 27/47		-	-	-
	SM570	460	450	430	570-720	19-26	-	-	47	-	-

8) Plate General Informations (ASTM/EN/JIS/ABS/LR)

Main Standards and Specifications

Standards		Specifications
ABS Specifications (American Bureau of Shipping Rules)	ASS (1998):	Rules Requirements for Materials and Welding, Part 2. Ordinary strength hull **structural steel** plates. Higher strength hull **structural steel** plates.
ASTM Specifications (American Society for Testing of Materials)	A36 (1996): A283 (1993): A285 (1990): A516 (1990): A572 (1997):	Standard specification for Carbon **Structural Steel**. Standard specification for Low and Intermediate Tensile Strength Carbon Steel Plates Standard specification for **Pressure Vessel** Plates, Carbon Steel, Low- and Intermediate Tensile Strength. Standard specification for **Pressure Vessel** Plates, Carbon Steel, for Moderate- and Lower-Temperature Service. Standard specification for High-Strength, Low Alloy, Columbium-Vanadium **Structural Steel**.
BSI Specifications (British Standards Institute)	BS 7191 (1989): EN 10028 (1993): EN 10113 (1993): EN 10137 (1996): EN 10149 (1996): EN 10207 (1992):	Specification for **Weldable structural steels** for fixed offshore structures. Specification for Flat products made of steels for **pressure purposes**. Standard specification for Hot rolled products in weldable fine grain **structural steels**. Plates and wide flats made of high yield strength **structural steels** in the quenched and tempered or precipitation hardened conditions. Specification for Hot-rolled flat products made of high yield strength steels for **cold forming**. Steels for simple **pressure vessels** - Technical delivery requirements for plates, strips and bars.
JIS Specifications (Japanese Industrial Standards)	G 3101 (1991): G 3106 (1995):	Rolled steels for **general structures.** Rolled steels for **welded structure**.
LRS Specifications (Lloyd's Register of Shipping Rules)	LR (1998):	Manufacture, Testing and Certification of Materials.